Cheatsheet AZ-104 — 14 · DNS and load balancing: Azure DNS, private zones, Load Balancer and its troubleshooting

1. Azure DNS (public zones)

Concept	Detail
Zone	Hosts the records of a domain you already **own** (Azure DNS **does not sell** domains: that is **App Service Domain**). On creation it receives 4 name servers (`ns1-xx.azure-dns.com / .net / .org / .info`): you must **delegate** at the registrar by changing the domain's **NS** records. Several zones with the same name can coexist (in different RGs); only the delegated one answers
Records	A, AAAA, **CNAME**, MX, NS, PTR, SOA, SRV, TXT, CAA; record set = same name and type; **TTL** per record set. A **CNAME** cannot coexist with other types under the same name nor sit at the **apex** (`@`)
Alias records	A / AAAA / CNAME pointing to an **Azure resource** (public IP, Traffic Manager profile, Front Door / CDN endpoint) or to another record set in the zone; they update automatically if the IP changes and **are valid at the apex** ("contoso.com → Traffic Manager")
Subdomains	Child zone (`dev.contoso.com`) delegated with **NS** records in the parent zone
Import / export	BIND zone files: `az network dns zone import / export`
Permissions	**DNS Zone Contributor**; RBAC per zone or per record set
DNSSEC	Zone signing (DS in the parent)
Verify	`nslookup -type=SOA contoso.com` · `nslookup www.contoso.com ns1-01.azure-dns.com` · `dig`

2. Private resolution

Mechanism	What it resolves	Detail
Azure-provided DNS (default, 168.63.129.16)	Names of VMs in the **same VNet** (`internal.cloudapp.net` suffix) and public names	Does not cross VNets; cannot be customized
Private DNS zone	Your own names (`contoso.internal`, `privatelink.*`) for one or several VNets	It is **linked** to each VNet: link with **autoregistration** (the VNet is a *registration* VNet: Azure creates and deletes the **A** records of its VMs; **only one registration zone per VNet**; up to 100 per zone) or **resolution** only (up to 1,000 per zone); a single link per zone-VNet pair; also **reverse** zones (PTR) and **split-horizon** (same zone public and private with different records)
Custom DNS (on the VNet or NIC)	Whatever your server decides (DC, BIND)	To resolve AD / on-prem; the server must **forward** to 168.63.129.16 for Azure names; VMs need a restart after changing the setting
Azure DNS Private Resolver	**VM-less** bridge between on-prem and Azure: **inbound endpoint** (IP in your VNet to which on-prem forwards the queries for private / privatelink zones) and **outbound endpoint + ruleset** (Azure forwards specific domains to the on-prem DNS servers)	Replaces VM-based forwarders; linked to VNets like a zone

- "VMs in VNet A must resolve VMs in VNet B by name" → a Private DNS zone linked to both (registration on those that create VMs) **or** custom DNS; peering only provides IP, not names.
- "On-premises clients must resolve the private endpoint of the storage account" → Private Resolver **inbound** + conditional forwarding of `privatelink.blob.core.windows.net` from the on-prem DNS servers.
- "Azure VMs must resolve contoso.local on-premises" → **outbound** endpoint with ruleset `contoso.local → on-prem DNS IP` (or custom DNS pointing to the DC).

3. Azure Load Balancer (layer 4)

Component	Detail
SKU	**Standard**: backend any VM / VMSS in a **single VNet** (by NIC or by IP), **zone-redundant**, **99.99 %** SLA, TCP / HTTP / **HTTPS** probes, **HA ports**, **outbound rules**, multiple frontends, TCP reset, **secure by default** (inbound traffic requires an **NSG** that allows it), global peering and Private Link; **Standard** public IP mandatory. **Basic**: **retired on September 30, 2025** (no SLA, availability set / VMSS only, open by default). **Gateway**: for NVA chains
Public vs internal	Public: frontend with a public IP (internet → VMs) · **Internal (ILB)**: frontend with a **private IP** from a subnet (web tier → app tier, on-prem → Azure)
Backend pool	VM / VMSS NICs or **IP addresses** in the same VNet; **no** availability set needed; VMs across several zones
Health probe	TCP (handshake) · HTTP / HTTPS (GET to a path; only **200** = healthy); default interval **5 s** in the portal (15 s via API), failure threshold; source **168.63.129.16** (AzureLoadBalancer tag); if **all** fail, Standard keeps the existing TCP flows but sends no new ones
Load balancing rule	Frontend IP:port → backend port, **TCP / UDP** protocol, probe, **session persistence** (None = 5-tuple hash · **Client IP** · Client IP and protocol), **idle timeout** 4-100 min (default 4), **floating IP** (Direct Server Return: for **SQL Always On listener** and NVAs), TCP reset
HA ports (ILB only)	One rule for **all ports** (and protocols): highly available NVAs
Inbound NAT rule	Port forwarding to **one** specific VM (e.g. frontend 50001 → VM1:3389, 50002 → VM2:3389); no probe required
Outbound rule	Explicit SNAT so the backend reaches the internet through the LB's public IP (**public** Standard LB only); a Standard ILB **provides no outbound access**: NAT Gateway (recommended), PIP or public LB with outbound rule
Global tier	Cross-region LB (public) that distributes across regional LBs

- It is **layer 4**: it does not terminate TLS, does not route by URL or host, does not inspect. For that, **Application Gateway**.

4. Choosing the load balancer

Service	Layer	Scope	Purpose	Signal
Load Balancer	4 (TCP / UDP)	Regional	VMs / VMSS, non-HTTP traffic, minimum latency, ILB between tiers, SQL listener	"TCP", "UDP", "internal", "SQL Always On", "RDP port forwarding"
Application Gateway	7 (HTTP / HTTPS)	Regional	**WAF**, TLS termination, **URL / host-based routing** (multi-site), cookie affinity, redirection, rewrite, autoscale v2, VM / VMSS / App Service / IP backends; dedicated subnet (/24 recommended)	"web application firewall", "/images to pool A", "multiple sites one IP", "SSL offload"
Front Door	7 global	Global (anycast)	Global entry point for multi-region web apps with **CDN**, WAF, TLS, fast failover, caching	"global", "lowest latency worldwide", "web app in two regions"
Traffic Manager	DNS	Global	Returns the endpoint IP according to the method: **Priority** (failover), **Weighted**, **Performance** (latency), **Geographic**, Multivalue, Subnet; Azure, **external** or nested endpoints; does not proxy traffic, DNS TTL	"DNS-based", "failover to another region", "route by user location", "any protocol"

- "HTTP + regional" → App Gateway · "HTTP + global" → Front Door · "non-HTTP + regional" → Load Balancer · "non-HTTP + global" or external endpoints → Traffic Manager. They combine: Traffic Manager / Front Door in front of regional App Gateways or LBs.

5. Load Balancer troubleshooting

Symptom	Usual cause	Fix
All backends **unhealthy**	The app is not listening on the probe port, the **NSG** blocks AzureLoadBalancer, the OS firewall, or the HTTP path does not return **200**	Check the listener, allow 168.63.129.16, adjust the probe (TCP vs HTTP and path)
Public LB unreachable from the internet	Standard LB is **secure by default**: NSG rule missing on subnet / NIC; public IP SKU does not match	Allow in NSG; Standard IP with Standard LB
The same VM always receives the traffic	**Session persistence** Client IP, or few clients (hash)	Set None if affinity is not needed
VM behind an ILB has no internet	Standard ILB provides no SNAT	NAT Gateway / outbound rule of a public LB / PIP
VM from another VNet cannot be added	Backend from **one VNet** only	Peering is not enough; move it or use App Gateway by IP
RDP to each VM on a different port	—	Inbound NAT rules
SQL listener does not respond	**Floating IP** missing on the rule	Enable floating IP and configure the listener
Asymmetric traffic, timeouts with NVA	UDRs without HA ports / without symmetric flows	HA ports + consistent routes
SNAT port exhaustion	Many outbound connections through the same IP	NAT Gateway (64,512 ports per IP, several IPs)
Metrics	**Health probe status**, **Data path availability**, SNAT connection count	Monitor → Metrics of the LB

6. Commands that show up

Task	PowerShell (Az.Network / Az.Dns)	Azure CLI
Zone and records	`New-AzDnsZone -Name contoso.com -ResourceGroupName RG1` · `New-AzDnsRecordSet -Name www -RecordType A -ZoneName contoso.com -ResourceGroupName RG1 -Ttl 3600 -DnsRecords (New-AzDnsRecordConfig -IPv4Address 203.0.113.10)` · alias: `-TargetResourceId <pip id>` · `Get-AzDnsRecordSet`	`az network dns zone create -g RG1 -n contoso.com` · `az network dns record-set a add-record -g RG1 -z contoso.com -n www -a 203.0.113.10` · `az network dns record-set a create -g RG1 -z contoso.com -n @ --target-resource <pip id>` · `az network dns record-set cname set-record -g RG1 -z contoso.com -n app -c app1.azurewebsites.net` · `az network dns zone show -g RG1 -n contoso.com --query nameServers`
Private zone	`New-AzPrivateDnsZone -Name contoso.internal -ResourceGroupName RG1` · `New-AzPrivateDnsVirtualNetworkLink -ZoneName contoso.internal -ResourceGroupName RG1 -Name link1 -VirtualNetworkId <id> -EnableRegistration` · `New-AzPrivateDnsRecordSet -Name db -RecordType A -ZoneName contoso.internal -ResourceGroupName RG1 -Ttl 300 -PrivateDnsRecords (New-AzPrivateDnsRecordConfig -IPv4Address 10.0.1.4)`	`az network private-dns zone create -g RG1 -n contoso.internal` · `az network private-dns link vnet create -g RG1 -z contoso.internal -n link1 -v vnet1 --registration-enabled true` · `az network private-dns record-set a add-record -g RG1 -z contoso.internal -n db -a 10.0.1.4`
VNet DNS	`$vnet.DhcpOptions.DnsServers = @("10.0.0.4"); Set-AzVirtualNetwork`	`az network vnet update -g RG1 -n vnet1 --dns-servers 10.0.0.4 10.0.0.5` (back to Azure: `--dns-servers null`)
Private Resolver	`New-AzDnsResolver` · `New-AzDnsResolverInboundEndpoint` · `New-AzDnsResolverOutboundEndpoint` · `New-AzDnsForwardingRuleset` · `New-AzDnsForwardingRulesetForwardingRule`	`az dns-resolver create` · `az dns-resolver inbound-endpoint create` · `az dns-resolver outbound-endpoint create` · `az dns-resolver forwarding-ruleset create` · `az dns-resolver forwarding-rule create --domain-name contoso.local. --target-dns-servers '[{"ip-address":"192.168.1.10"}]'`

Task	PowerShell (Az.Network / Az.Dns)	Azure CLI
Load Balancer	`New-AzLoadBalancer -Name lb1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -FrontendIpConfiguration $fe -BackendAddressPool $be -Probe $probe -LoadBalancingRule $rule` with `New-AzLoadBalancerFrontendIpConfig`, `New-AzLoadBalancerBackendAddressPoolConfig`, `New-AzLoadBalancerProbeConfig -Protocol Http -Port 80 -RequestPath / -IntervalInSeconds 5 -ProbeCount 2`, `New-AzLoadBalancerRuleConfig -Protocol Tcp -FrontendPort 80 -BackendPort 80 -EnableFloatingIP -LoadDistribution SourceIP` · NAT: `Add-AzLoadBalancerInboundNatRuleConfig` · add a NIC: `$nic.IpConfigurations[0].LoadBalancerBackendAddressPools.Add($pool); Set-AzNetworkInterface`	`az network lb create -g RG1 -n lb1 --sku Standard --public-ip-address pip-lb --frontend-ip-name fe --backend-pool-name be` (internal: `--vnet-name vnet1 --subnet sub1 --private-ip-address 10.0.1.100`) · `az network lb probe create -g RG1 --lb-name lb1 -n p80 --protocol Http --port 80 --path / --interval 5 --threshold 2` · `az network lb rule create -g RG1 --lb-name lb1 -n r80 --protocol Tcp --frontend-port 80 --backend-port 80 --frontend-ip-name fe --backend-pool-name be --probe-name p80 --load-distribution SourceIP --floating-ip true --idle-timeout 15` · `az network lb inbound-nat-rule create -g RG1 --lb-name lb1 -n rdp1 --protocol Tcp --frontend-port 50001 --backend-port 3389 --frontend-ip-name fe` · `az network nic ip-config address-pool add -g RG1 --nic-name nic1 -n ipconfig1 --lb-name lb1 --address-pool be` · `az network lb outbound-rule create`
App Gateway / Traffic Manager (quick)	`New-AzApplicationGateway` · `New-AzTrafficManagerProfile -TrafficRoutingMethod Priority -RelativeDnsName contoso -Ttl 30 -MonitorProtocol HTTPS -MonitorPort 443 -MonitorPath /` + `New-AzTrafficManagerEndpoint`	`az network application-gateway create --sku WAF_v2 ...` · `az network traffic-manager profile create -n tm1 -g RG1 --routing-method Performance --unique-dns-name contoso` · `az network traffic-manager endpoint create --type azureEndpoints --target-resource-id <pip id>`

7. Decision tree

```
What does the wording ask for?
├─ Host a public domain I already own ..... Azure DNS zone + delegate NS at the registrar
├─ Buy a domain from Azure ..... App Service Domain
├─ contoso.com (apex) → Traffic Manager / public IP ..... Alias record (not CNAME)
├─ Private names across several VNets ..... Private DNS zone linked to all the VNets
├─ On-prem resolves private zones / private endpoints ..... DNS Private Resolver inbound + on-prem conditional forwarding
├─ Azure resolves on-prem domains ..... Private Resolver outbound + ruleset (or custom DNS)
├─ Distribute TCP / UDP across VMs in one VNet ..... Standard Load Balancer (public or internal)
├─ RDP to each VM on different ports of one IP ..... Inbound NAT rules
├─ SQL Always On listener ..... ILB + floating IP
├─ NVAs in HA ..... ILB + HA ports
├─ Internet egress for VMs behind an ILB ..... NAT Gateway (or public LB outbound rule)
├─ WAF, TLS offload, URL routing, multiple sites ..... Application Gateway
├─ Global multi-region web with CDN and WAF ..... Front Door
└─ DNS-based failover between regions, any protocol ..... Traffic Manager (Priority)
```

8. Signals → answer

Wording	Answer
"DNS zone created but the domain does not resolve"	NS delegation at the registrar is missing
"point the root domain to the web app / Traffic Manager"	Alias record A / CNAME-alias (CNAME not valid at the apex)
"CNAME and TXT with the same name"	Not allowed: CNAME cannot coexist with other types
"VMs register their names automatically"	Private zone with autoregistration link (one per VNet)
"VNet already has a registration zone"	Only one registration zone per VNet: the second one is resolution only
"private endpoint resolves to public IP from on-premises"	Private Resolver inbound + conditional forwarder
"replace the DNS forwarder VMs"	DNS Private Resolver
"Basic load balancer"	Retired: Standard
"load balancer across VMs in two VNets"	No: one VNet per backend pool
"VMs must be in an availability set to use the LB"	Only with Basic (retired); Standard does not require it
"backend shows unhealthy"	Probe: port, NSG (AzureLoadBalancer), HTTP 200
"requests from the same client must go to the same VM"	Session persistence Client IP
"connections drop after 4 minutes idle"	Raise the idle timeout (up to 100 min) or keep-alive
"VMs behind the internal LB cannot reach the internet"	NAT Gateway / outbound rule
"SSL termination and URL routing"	Application Gateway
"protect the web app with WAF"	Application Gateway WAF_v2 or Front Door Premium
"route users to the closest region"	Traffic Manager Performance (or Front Door for HTTP)
"failover to secondary region only when primary fails"	Traffic Manager Priority
"send 10 % of traffic to the new deployment"	Traffic Manager Weighted (DNS) or App Service slots

9. Mistakes to avoid

1. Forgetting the NS delegation or placing a CNAME at the apex.

2. Assuming peering resolves names: a Private DNS zone or custom DNS is needed.
3. Linking two zones with autoregistration to the same VNet.
4. Choosing Load Balancer for WAF, TLS or URL routing, or Application Gateway for UDP.
5. Designing with Basic LB or Basic IP (retired) or mixing IP and LB SKUs.
6. Blocking the probe (168.63.129.16) with the NSG or probing a port where nothing listens.
7. Expecting a Standard ILB to provide internet egress.
8. Forgetting floating IP for SQL Always On or HA ports for NVAs.